

Tutorial-5.

24/2/26.

SVM.

Class (SPam = +1, Not SPam = -1)

Suspicious (w ₁) Words	Links (w ₂)	Attachment (w ₃)	(Bias: w ₀) Class(y)
6	3	1	+1
1	0	0	-1
4	2	1	+1
0	0	0	-1
5	2	1	+1

Weighted sum = $z = w_0 + w_1x_1 + w_2x_2 + w_3x_3$.

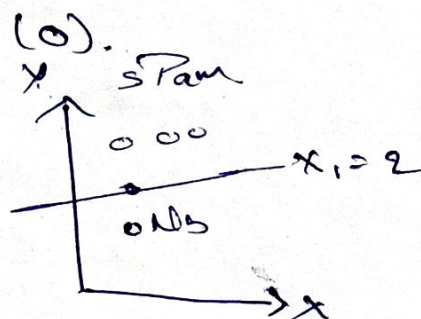
SPam $\rightarrow$ $\begin{matrix} x_1 \\ (4, 5, 6) \end{matrix}$ $\begin{matrix} x_2 \\ (2, 3) \end{matrix}$ $\begin{matrix} x_3 \\ (1) \end{matrix}$

Non-SPam $\rightarrow$ $\begin{matrix} (0, 1) \end{matrix}$ $\begin{matrix} (0) \end{matrix}$ $\begin{matrix} (0) \end{matrix}$

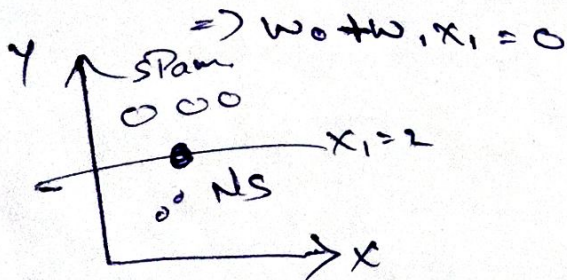
Initialize: $w_1 \& w_3 = 0$

SPam. $f(x) = 0$.

non-SPam $\downarrow$
Decision Boundary.



If $f(x) = 0 \Rightarrow w_0 + w_1x_1 + w_2x_2 + w_3x_3 = 0$.



Now; $w_1 = 1$.

$$w_0 + w_1x_1 = 0$$

$$w_0 + (1 \cdot 2) = 0$$

$$\boxed{w_0 = -2}$$

$$f(x) = w_1 x_1 + w_2 x_2 + w_3 x_3 = -2.$$

calculate

(8)

check

$$y_i(x_1 w_1 + x_2 w_2 + x_3 w_3 + w_0) \geq 1 \rightarrow \text{Perfect classification}$$

$$y_i(x_1 w_1 + x_2 w_2 + x_3 w_3 + w_0) \leq 1 \rightarrow \text{classification fail}$$

$$\begin{aligned} ① &= 1(6(1) + 3(0) + 1(0) + (-2)) \quad \text{check } y_i(x_i - 2) \\ &= 1(6 - 2) \\ &= 4 > 1 \end{aligned}$$

$$\begin{aligned} ② &= -(1 - 2) \\ &= 1 \end{aligned} \quad \begin{aligned} &\geq 1 \text{ (P.F)} \\ &\geq 1 \text{ (P.F)} \end{aligned}$$

$$\begin{aligned} ③ &= 1(4 - 2) \\ &= 2 \end{aligned} \quad \begin{aligned} &\geq 1 \text{ (P.F)} \\ &\geq 1 \text{ (P.F)} \end{aligned}$$

$$\begin{aligned} ④ &= -1(0 - 2) \\ &= 2 \end{aligned} \quad \geq 1 \text{ (P.F)}$$

classification error
hyperparameter.

$$\Rightarrow L = \frac{1}{2} \|w\|^2 - \sum_i c_i [y_i (w x_i + w_0) - 1]$$

$$= \frac{1}{2} |w_1^2 + w_2^2 + w_3^2| - \sum_{i=1}^N c_i \text{ error } y_i - \hat{y}_i$$

Update weights

$$\text{If P.F} \Rightarrow y_i (w_1 x_1 + w_2 x_2 + \dots) \geq 1$$

$\rightarrow 0$

$$\rightarrow L = \frac{1}{2} w^2$$

$$w^{\text{New}} = w^{\text{old}} - \eta \frac{\partial L}{\partial w}$$

$$= w - \eta w$$

$$w_i^{\text{New}} = w_i^{\text{old}} - \eta w_i^{\text{old}}$$

$$w_0^{\text{New}} = w_0^{\text{old}} - \eta w_0^{\text{old}}$$

$$\eta = 0.1$$

$$w_0^{\text{New}} = 1 - (0.1 \times 1)$$

$$= 1 - 0.1$$

$$= 0.9.$$

$$\text{Margin} = \frac{2}{\|w\|} = \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^2 + \dots}}$$

$$\Rightarrow \text{IF N.T.F} \Rightarrow \forall_i (w_1 x_1 + w_2 x_2 + \dots) < 1$$

$$w^{\text{new}} = w - \eta \frac{\partial L}{\partial w}$$

$$= w - \eta (w - \sum_i C_i y_i x_i)$$

$$= (1 - \eta) w + \eta \sum C_i y_i x_i$$

$$\sum C_i y_i x_i = C \sum y_i x_i$$

$$C = 1 \Rightarrow (y_1 x_{11} + y_2 x_{12} + y_3 x_{13} + y_4 x_{14} + y_5 x_{15})$$

SVM

Weighted sum: $f(x) = z = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$

Goal: check: $y: [f(x)] \geq 1 \rightarrow$ Perfect classification.
 $< 1 \Rightarrow$ Classification fails

Initialize $w_1 = 1$

$$w_2 = 1$$

$$w_3 = 0.1 < 0 = (1 + 0.1) = 1.1$$

$$b = 0$$

1st set of inputs: $y_1(f(x)) = -0 < 1$

$$w_j^{\text{new}} = w_j^{\text{old}} - \eta \frac{\partial L}{\partial w}$$

$$w_j^{\text{new}} = w_j^{\text{old}} - \eta \left(w = \sum_i c_i y_i x_{ij} \right)$$

$$w_1^{\text{new}} = w_1^{\text{old}} - \eta [w_1^{\text{old}} - \sum y_i x_{i1}] \quad \boxed{c=1}$$

$$w_1^{\text{new}} = w_1^{\text{old}} - \eta \frac{\partial L}{\partial w}$$

$$w_1^{\text{new}} = 1 - 1 [1 - (-1 \times 6)] = -6$$

$(x_1) \quad (x_2) \quad (x_3) \quad (y)$

6 3 1 +1

1 0 0 -1

4 2 1 +1

0 0 0 -1

5 2 1 +1

$$w_2^{\text{new}} = 1 - 1 [1 - (-1 \times 3)] = -3$$

$$w_3^{\text{new}} = 1 - 1 [1 - (-1 \times 1)] = -1$$

$$b^{\text{new}} = b^{\text{old}} - \eta \frac{\partial L}{\partial b}$$

$$= b^{\text{old}} - \eta \left[\sum c_i y_i \right]$$

$$n = 1$$

$$b = 0$$

$$c = 1$$

$$y = 1$$

$$= 0 - [(-1)] = 1$$

for Perfect

2nd set of inputs.

$$w_1 = -6$$

$$w_2 = -3$$

$$w_3 = -1$$

$$b = 1$$

$$y_i \times f(x) = -1(-6+1) = 5 > 1$$

If Right Prediction: $\rightarrow$ no need to add error then.

$$\frac{\partial L}{\partial w} = w$$

$$w_j^{\text{new}} = w_j^{\text{old}} - \eta \frac{\partial L}{\partial w}$$

$$w_j^{\text{new}} = w_j^{\text{old}} - \eta w_j^{\text{old}}$$

$$w_1^{\text{new}} = -6 - (1-6) = 0$$

$$w_2^{\text{new}} = -3 - (1-3) = 0$$

$$w_3^{\text{new}} = -1 - (1-1) = 0$$

so take η as 0.1

① Calculate the margin:

$$\text{Margin} = \frac{2}{\|w\|} = \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^2}}$$

$$b^{\text{new}} = b^{\text{old}} - \eta \left(\frac{\partial L}{\partial b} \right)$$

$$= b^{\text{old}} - \eta (\sum c_i y_i) - 1$$

$$= 1 - [0] - [(1 \times 1) - 1] = 1$$

$$b^{\text{new}} = 1$$